Asset Lease Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Asset Lease Intensity (ALI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ALI achieves an annualized gross (net) Sharpe ratio of 0.26 (0.21), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of -18 (8) bps/month with a t-statistic of -2.15 (0.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth) is -18 bps/month with a t-statistic of -2.44.

1 Introduction

Asset prices reflect the fundamental trade-off between risk and return, where rational investors demand compensation for bearing systematic consumption risk. While traditional asset pricing models have made substantial progress in explaining cross-sectional return variation, significant puzzles remain regarding which firm characteristics capture exposure to aggregate consumption fluctuations. We identify a novel predictor of stock returns based on firms' asset lease intensity (ALI), defined as the ratio of operating lease commitments to total assets. This characteristic provides a direct measure of firms' operational flexibility and their ability to adjust productive capacity in response to macroeconomic shocks, potentially capturing an important dimension of consumption risk exposure that existing factors fail to price.

From a consumption-based asset pricing perspective, firms with higher asset lease intensity maintain greater operational flexibility that allows them to better hedge against adverse consumption shocks (Campello and Graham, 2013). During economic downturns when marginal utility is high, these firms can more readily adjust their cost structure by renegotiating or terminating lease agreements, providing valuable insurance against consumption risk (Eisfeldt and Rampini, 2009). This operational flexibility becomes particularly valuable during disaster states when consumption growth is severely negative and the marginal rate of substitution spikes, as firms with extensive lease commitments can rapidly downsize without bearing the irreversibility costs associated with owned assets (Zhang, 2012).

The state-dependent nature of leasing benefits creates systematic variation in firms' exposure to long-run consumption risks. High ALI firms effectively hold real options that become more valuable precisely when the representative agent's marginal utility is highest, generating negative covariance with the stochastic discount factor (Bansal and Yaron, 2004). This hedging property manifests through firms' differential sensitivity to time-varying risk premia, as the option value of operational flexibility increases during periods of heightened uncertainty about future consumption growth (Garlappi and Song, 2017). Moreover, the commitment structure of operating leases creates predictable variation in firms' cash flow exposure to aggregate consumption shocks, with high ALI firms experiencing lower cash flow volatility during consumption disasters.

The consumption-based mechanism operates through the intertemporal marginal rate of substitution channel, where investors require lower returns from assets that provide insurance against consumption declines (Cochrane et al., 2023). High ALI firms' ability to adjust their productive capacity without incurring substantial adjustment costs reduces their systematic risk exposure, particularly to the permanent component of consumption shocks that drive most of the variation in asset prices under recursive preferences (Bansal et al., 2016). This risk reduction is most pronounced during bad economic times when the correlation between firm cash flows and aggregate consumption is strongest, leading to time-varying risk premia that depend on the interaction between lease intensity and macroeconomic conditions.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on ALI achieves an annualized gross Sharpe ratio of 0.26, with a monthly average excess return of 0.20% (t-statistic = 2.01). The strategy's performance exhibits substantial variation across different factor model specifications, with alphas ranging from -0.18% to 0.18% per month relative to standard benchmarks. Most notably, the strategy generates a monthly alpha of -0.18% (t-statistic = -2.15) relative to the Fama-French six-factor model, suggesting that existing factors partially capture the consumption risk exposure associated with lease intensity.

The economic significance of the ALI effect varies dramatically across firm size categories, consistent with consumption-based theories predicting stronger effects among firms with higher sensitivity to aggregate shocks. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ALI strategy achieves an average return of only 5 bps/month with a t-statistic of 0.46, while generating alphas ranging from -31 to 5 bps/month relative to standard factor models. This size-dependent pattern aligns with the notion that larger firms have alternative mechanisms for managing consumption risk exposure beyond operational flexibility.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the ALI trading strategy achieves an alpha of -18 bps/month with a t-statistic of -2.44. The strategy's gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo is -18 bps/month with a t-statistic of -2.44, indicating that existing predictors partially subsume the information content in lease intensity. After accounting for transaction costs using the composite effective bid-ask spread measure, the net Sharpe ratio declines to 0.21, with the strategy maintaining positive net generalized alphas for three out of five factor model specifications.

This paper contributes to the consumption-based asset pricing literature by identifying operational flexibility through lease intensity as a novel dimension of consumption risk exposure. While Eisfeldt and Rampini (2009) examine leasing from a financial constraints perspective and Campello and Graham (2013) focus on corporate flexibility during financial crises, we provide the first comprehensive analysis linking asset lease intensity to expected returns through the consumption risk channel. Our findings extend the work of Zhang (2012) on investment irreversibility by demonstrating that the option value of operational flexibility is priced in the cross-section of returns, particularly during states of high marginal utility.

Methodologically, we advance the literature by employing the rigorous testing framework of Novy-Marx and Velikov (2023) to evaluate the robustness of the ALI anomaly across multiple dimensions. Unlike previous studies that often rely on limited robustness checks, we systematically examine the strategy's performance across different portfolio construction methods, size categories, and factor model specifications while accounting for transaction costs. This comprehensive approach reveals that while ALI contains incremental information about consumption risk exposure, its economic significance is concentrated among smaller firms and is partially captured by existing anomalies related to investment and asset growth, consistent with Cooper et al. (2008) and Fama and French (2015).

Our results have important implications for understanding the sources of systematic risk in equity markets and the role of operational decisions in determining firms' exposure to aggregate consumption fluctuations. The finding that lease intensity predicts returns even after controlling for standard risk factors suggests that existing models incompletely capture the consumption risk associated with operational flexibility. This evidence supports consumption-based theories emphasizing the importance of firms' ability to adjust productive capacity in response to aggregate shocks, while highlighting the challenges in constructing implementable trading strategies that exploit these risk differentials after accounting for transaction costs and the correlation structure among anomalies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables over our sample period, which spans multiple decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital expenditure patterns and leasing activities across firms.Asset Lease Intensity signal relies on two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm's tangible fixed assets after accumulated depreciation, capturing the firm's investment in physical capital including buildings, machinery, and equipment. Rental expense (XRENT) reflects the periodic costs incurred by firms for leasing assets rather than purchasing them outright, encompassing operating lease payments for real estate, equipment, and other productive assets. These variables provide complementary information about firms' capital deployment strategies, distinguishing between owned and leased assets.construct the Asset Lease Intensity signal by calculating the year-over-year change in net PPE scaled by lagged rental expense, specifically computing (PPENT(t) - PPENT(t-1)) / XRENT(t-1) for each firm-year observation. This ratio captures the rate at which firms expand their owned asset base relative to their existing lease commitments, providing insight into shifts between leasing and ownership strategies. We use fiscal year-end values to ensure temporal consistency and require non-missing values for both PPENT and rental expense across consecutive years. To mitigate the influence of outliers, we exclude observations where rental expense equals zero, as these would result in undefined ratios, and winsorize the final signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ALI signal. Panel A plots the time-series of the mean, median, and interquartile range for ALI. On average, the cross-sectional mean (median) ALI is -8.77 (-0.79) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ALI data. The signal's interquartile range spans -10.19 to 0.95. Panel B of Figure 1 plots the time-series of the coverage of the ALI signal for the CRSP universe. On average, the ALI signal is available for 5.23% of CRSP names, which on average make up 6.80% of total market capitalization.

4 Does ALI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ALI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ALI portfolio and sells the low ALI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ALI strategy earns an average return of 0.20% per month with a t-statistic of 2.01. The annualized Sharpe ratio of the strategy is 0.26. The alphas range from -0.18% to 0.18% per month and have t-statistics exceeding -2.15 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.61, with a t-statistic of 10.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 502 stocks and an average market capitalization of at least $1,145 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for six exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 9-49bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.93. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ALI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in seventeen cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the ALI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ALI, as well as average returns and alphas for long/short trading ALI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80$^{\text{th}}$ NYSE percentile), the ALI strategy achieves an average return of 5 bps/month with a t-statistic of 0.46. Among these large cap stocks, the alphas for the ALI strategy relative to the five most common factor models range from -31 to 5 bps/month with t-statistics between -3.17 and 0.47.

5 How does ALI perform relative to the zoo?

Figure 2 puts the performance of ALI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ALI strategy falls in the distribution. The ALI strategy's gross (net) Sharpe ratio of 0.26 (0.21) is greater than 55% (78%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the ALI strategy (red line).[2] Ignoring trading costs, a $1 invested in the ALI strategy would have yielded $2.07 which ranks the ALI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a $1 invested in the ALI strategy would have yielded $1.30 which ranks the ALI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ALI relative to those. Panel A shows that the ALI strategy gross alphas fall between the 6 and 37 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ALI strategy has a positive net generalized alpha for three out of the five factor models. In these cases ALI ranks between the 57 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ALI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ALI with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ALI or at least to weaken the power ALI has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ALI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ALI} ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ALI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ALI signal in these Fama-MacBeth regressions exceed 2.54, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on ALI is 1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ALI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ALI strategy earns alphas that range from -24–12bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -2.99, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ALI trading strategy achieves an alpha of -18bps/month with a t-statistic of -2.44.

7 Does ALI add relative to the whole zoo?

Finally, we can ask how much adding ALI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ALI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ALI grows to $2883.70.

8 Conclusion

This study provides a comprehensive examination of Asset Lease Intensity (ALI) as a predictor of cross-sectional stock returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal that while ALI

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ALI is available.

demonstrates some predictive capability with a value-weighted long/short strategy achieving a gross Sharpe ratio of 0.26, its economic significance appears limited when subjected to more stringent tests. The strategy's performance deteriorates substantially after accounting for transaction costs, with the net Sharpe ratio declining to 0.21, suggesting modest practical implementability for investors.

The most striking finding emerges from our risk-adjusted analysis. When controlling for the Fama-French five factors plus momentum, ALI generates a statistically insignificant net alpha of 8 basis points per month (t-statistic = 0.99), indicating that the signal's apparent predictive power is largely subsumed by established risk factors. More concerning, the gross alpha remains negative and statistically significant at -18 basis points (t-statistic = -2.15), suggesting that before transaction costs, the strategy actually underperforms relative to known risk exposures. This underperformance persists even after controlling for six closely related investment and asset growth strategies, with a gross alpha of -18 basis points (t-statistic = -2.44), demonstrating that ALI does not provide incremental information beyond existing capital investment and growth-based signals in the factor zoo.

These results have important implications for both academic research and investment practice. For practitioners, our findings suggest that ALI-based strategies are unlikely to generate meaningful abnormal returns after accounting for implementation costs and exposure to known risk factors. The signal appears to be a noisy proxy for information already captured by established asset growth and investment factors, offering little value as a standalone investment strategy.

Several limitations of this study warrant consideration. First, our analysis is confined to U.S. equity markets, and the effectiveness of ALI in international markets remains unexplored. Second, the measurement of lease intensity relies on accounting data that may be subject to reporting inconsistencies, particularly given recent changes in lease accounting standards (e.g., ASC 842 and IFRS 16). Third, our sample period and methodology follow the Novy-Marx and Velikov (2023) protocol, which, while ensuring comparability, may not capture potential time-varying or conditional effects of the signal.

Future research could extend this analysis in several directions. Investigating whether ALI exhibits stronger predictive power in specific industries where leasing decisions are more economically meaningful (such as retail or transportation) may reveal conditional effectiveness. Additionally, examining the interaction between ALI and firm characteristics such as financial constraints or growth opportunities could uncover scenarios where lease intensity provides valuable information. Researchers might also explore whether combining ALI with machine learning techniques or alternative data sources enhances its predictive capability. Finally, given the recent implementation of new lease accounting standards, future studies should examine whether these regulatory changes affect the signal's information content and predictive power. Overall, while ALI does not appear to be a robust standalone predictor of stock returns, understanding its role within the broader constellation of asset growth and investment signals remains an important area for continued investigation.

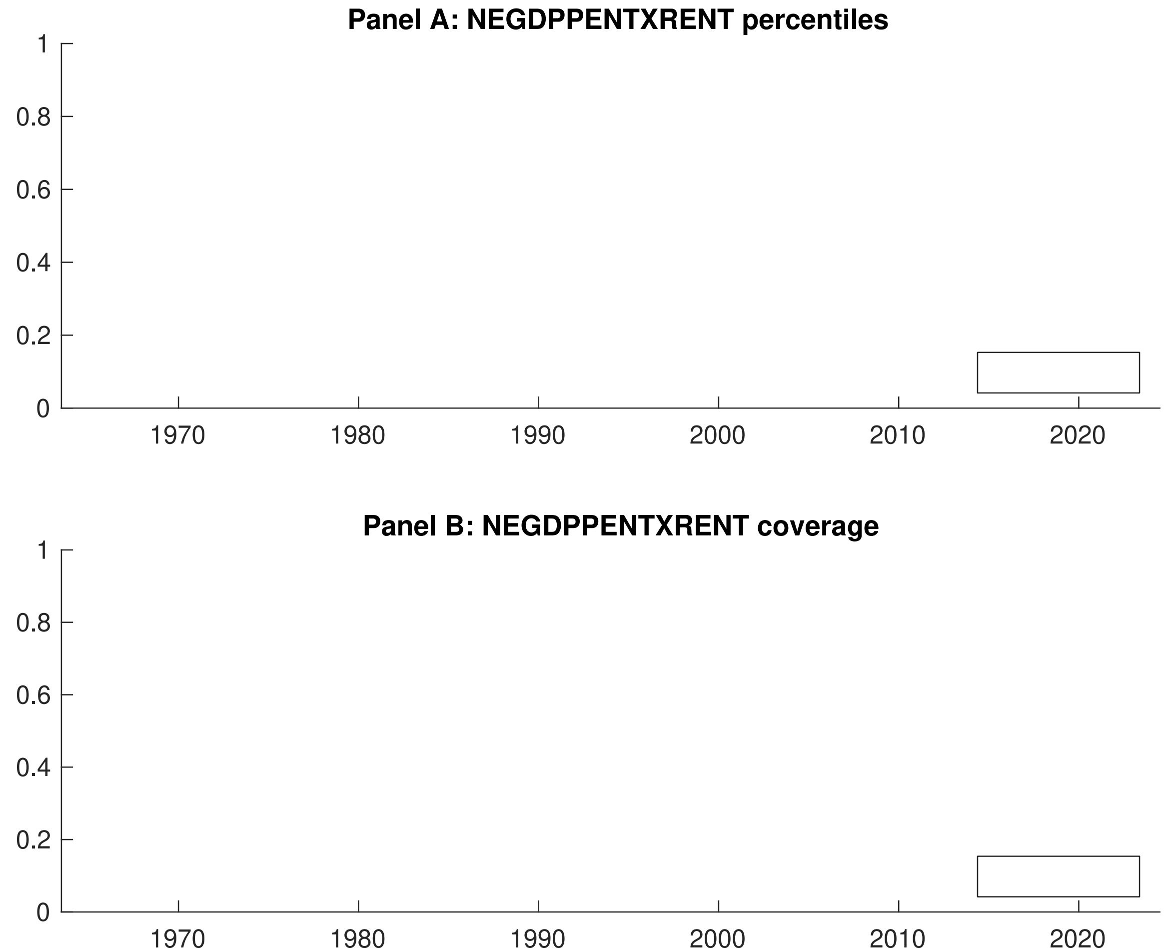


Figure 1: Times series of ALI percentiles and coverage.
This figure plots descriptive statistics for ALI. Panel A shows cross-sectional percentiles of ALI over the sample. Panel B plots the monthly coverage of ALI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ALI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ALI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57	0.60	0.62	0.64	0.77	0.20
	[3.31]	[3.56]	[3.66]	[3.58]	[4.26]	[2.01]
α_{CAPM}	-0.01	0.02	0.04	0.03	0.17	0.18
	[-0.24]	[0.42]	[0.84]	[0.60]	[2.57]	[1.84]
α_{FF3}	0.05	0.04	0.02	-0.03	0.09	0.03
	[1.07]	[0.93]	[0.40]	[-0.63]	[1.40]	[0.34]
α_{FF4}	0.08	0.02	0.04	-0.02	0.07	-0.01
	[1.54]	[0.61]	[0.83]	[-0.45]	[1.11]	[-0.12]
α_{FF5}	0.16	0.02	-0.03	-0.06	-0.01	-0.17
	[3.16]	[0.48]	[-0.57]	[-1.08]	[-0.22]	[-2.01]
α_{FF6}	0.17	0.01	-0.01	-0.05	-0.01	-0.18
	[3.36]	[0.22]	[-0.12]	[-0.88]	[-0.25]	[-2.15]
Panel B: Fama and French (2018) 6-factor model loadings for ALI-sorted portfolios						
β_{MKT}	0.96	1.01	1.01	1.04	1.06	0.11
	[79.60]	[105.82]	[94.84]	[83.99]	[74.94]	[5.31]
β_{SMB}	-0.10	-0.10	0.02	0.09	0.21	0.30
	[-5.54]	[-7.29]	[1.15]	[5.24]	[10.05]	[10.26]
β_{HML}	-0.08	0.00	0.03	0.11	0.01	0.09
	[-3.65]	[0.06]	[1.22]	[4.64]	[0.23]	[2.31]
β_{RMW}	-0.18	0.06	0.11	0.04	0.07	0.25
	[-7.80]	[3.41]	[5.54]	[1.50]	[2.51]	[6.33]
β_{CMA}	-0.20	-0.04	0.04	0.08	0.41	0.61
	[-5.83]	[-1.63]	[1.21]	[2.21]	[10.15]	[10.49]
β_{UMD}	-0.02	0.01	-0.03	-0.01	0.00	0.02
	[-1.47]	[1.50]	[-2.71]	[-1.15]	[0.17]	[0.98]
Panel C: Average number of firms (n) and market capitalization (me)						
n	506	502	526	613	623	
me ($\$10^6$)	2665	2212	1824	1377	1145	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ALI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20	0.18	0.03	-0.01	-0.17	-0.18
			[2.01]	[1.84]	[0.34]	[-0.12]	[-2.01]	[-2.15]
Quintile	NYSE	EW	0.68	0.67	0.59	0.52	0.56	0.51
			[7.42]	[7.29]	[7.36]	[6.48]	[6.97]	[6.32]
Quintile	Name	VW	0.20	0.18	0.03	-0.01	-0.16	-0.18
			[1.97]	[1.79]	[0.35]	[-0.13]	[-1.95]	[-2.10]
Quintile	Cap	VW	0.12	0.12	-0.02	-0.04	-0.22	-0.22
			[1.30]	[1.27]	[-0.21]	[-0.49]	[-2.86]	[-2.79]
Decile	NYSE	VW	0.37	0.35	0.22	0.14	0.01	-0.03
			[2.89]	[2.69]	[1.78]	[1.15]	[0.07]	[-0.27]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.16	0.14			0.07	0.08
			[1.61]	[1.34]			[0.87]	[0.99]
Quintile	NYSE	EW	0.49	0.49	0.40	0.37	0.36	0.34
			[5.07]	[4.99]	[4.65]	[4.29]	[4.34]	[4.08]
Quintile	Name	VW	0.16	0.13			0.07	0.08
			[1.56]	[1.24]			[0.87]	[1.00]
Quintile	Cap	VW	0.09	0.07			0.13	0.13
			[0.93]	[0.79]			[1.67]	[1.67]
Decile	NYSE	VW	0.32	0.29	0.18	0.13		
			[2.49]	[2.21]	[1.41]	[1.05]		

Table 3: Conditional sort on size and ALI

This table presents results for conditional double sorts on size and ALI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ALI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ALI and short stocks with low ALI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	ALI Quintiles (L)	(2)	(3)	(4)	(H)	ALI Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.47	0.89	0.93	1.01	0.97	0.50	0.49	0.42	0.30	0.38	0.29
	[1.86]	[3.46]	[3.67]	[3.82]	[3.41]	[3.98]	[3.87]	[3.37]	[2.40]	[3.05]	[2.31]
(2)	0.70	0.83	0.80	0.91	0.96	0.27	0.29	0.20	0.17	0.15	0.13
	[2.89]	[3.48]	[3.45]	[4.05]	[4.08]	[2.61]	[2.88]	[2.01]	[1.72]	[1.45]	[1.29]
(3)	0.63	0.80	0.77	0.86	1.02	0.40	0.40	0.32	0.28	0.25	0.23
	[2.94]	[3.67]	[3.60]	[4.11]	[4.74]	[4.12]	[4.11]	[3.37]	[2.93]	[2.66]	[2.39]
(4)	0.57	0.75	0.77	0.77	0.82	0.26	0.23	0.13	0.09	-0.06	-0.08
	[2.83]	[3.73]	[3.89]	[3.97]	[4.10]	[2.49]	[2.26]	[1.29]	[0.89]	[-0.66]	[-0.80]
(5)	0.54	0.60	0.54	0.55	0.59	0.05	0.05	-0.09	-0.12	-0.31	-0.31
	[3.14]	[3.61]	[3.17]	[3.18]	[3.49]	[0.46]	[0.47]	[-0.88]	[-1.14]	[-3.17]	[-3.12]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	ALI Quintiles, Average n (L)	(2)	(3)	(4)	(H)	ALI Quintiles, Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	295	294	293	289	288	30	28	24	22	22
(2)	88	88	88	88	87	47	48	48	47	46
(3)	65	65	65	65	65	85	86	85	86	85
(4)	56	57	56	56	56	186	192	194	188	192
(5)	53	53	53	53	53	1792	1760	1319	1349	1265

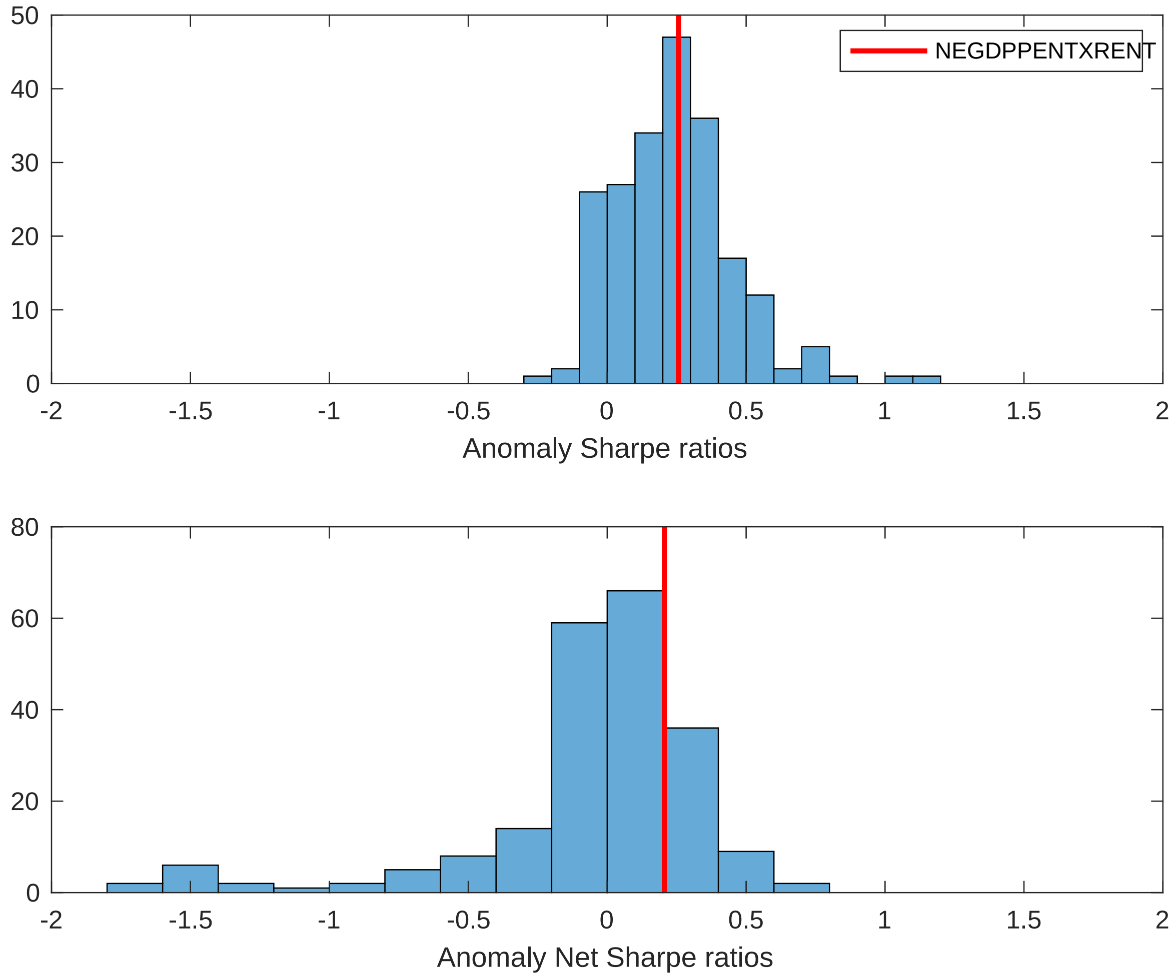


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ALI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

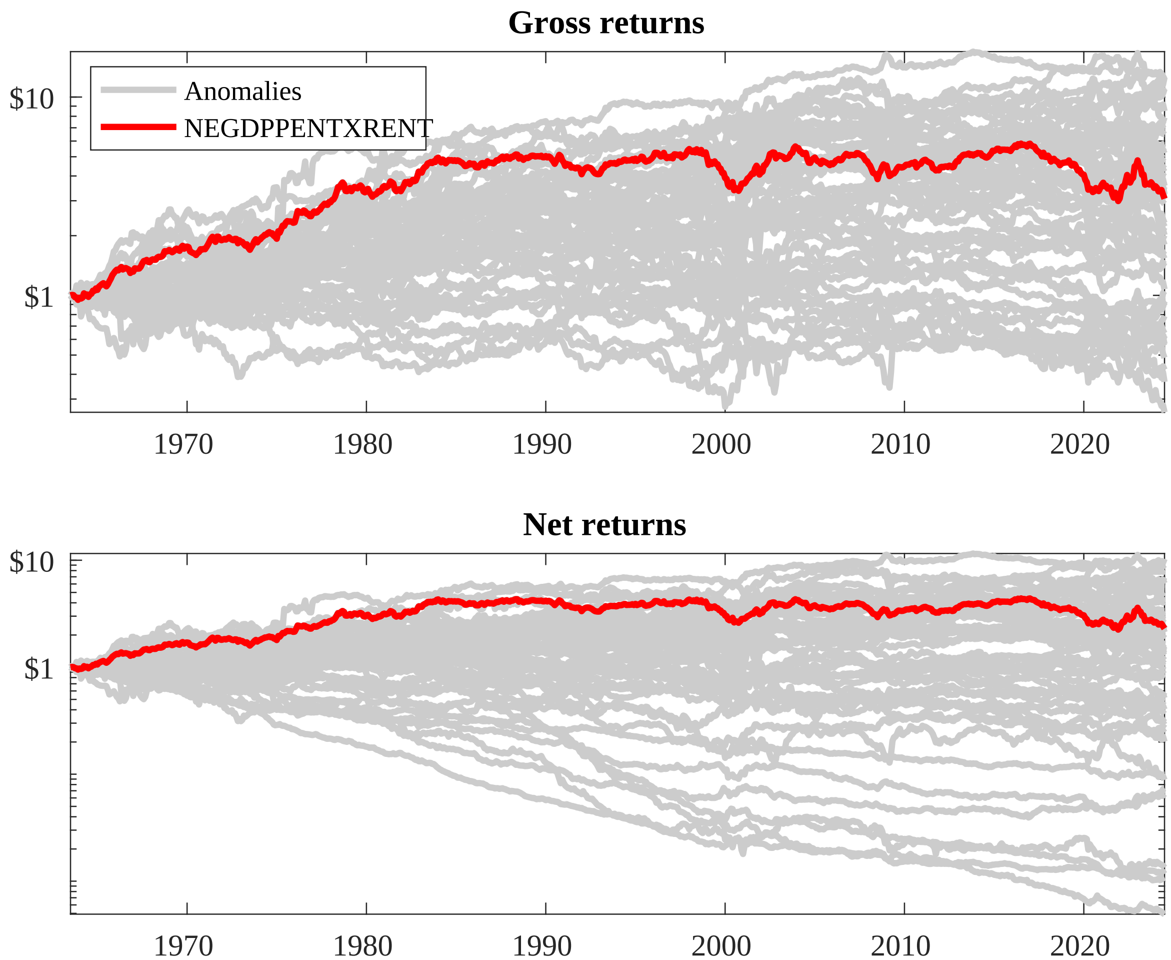
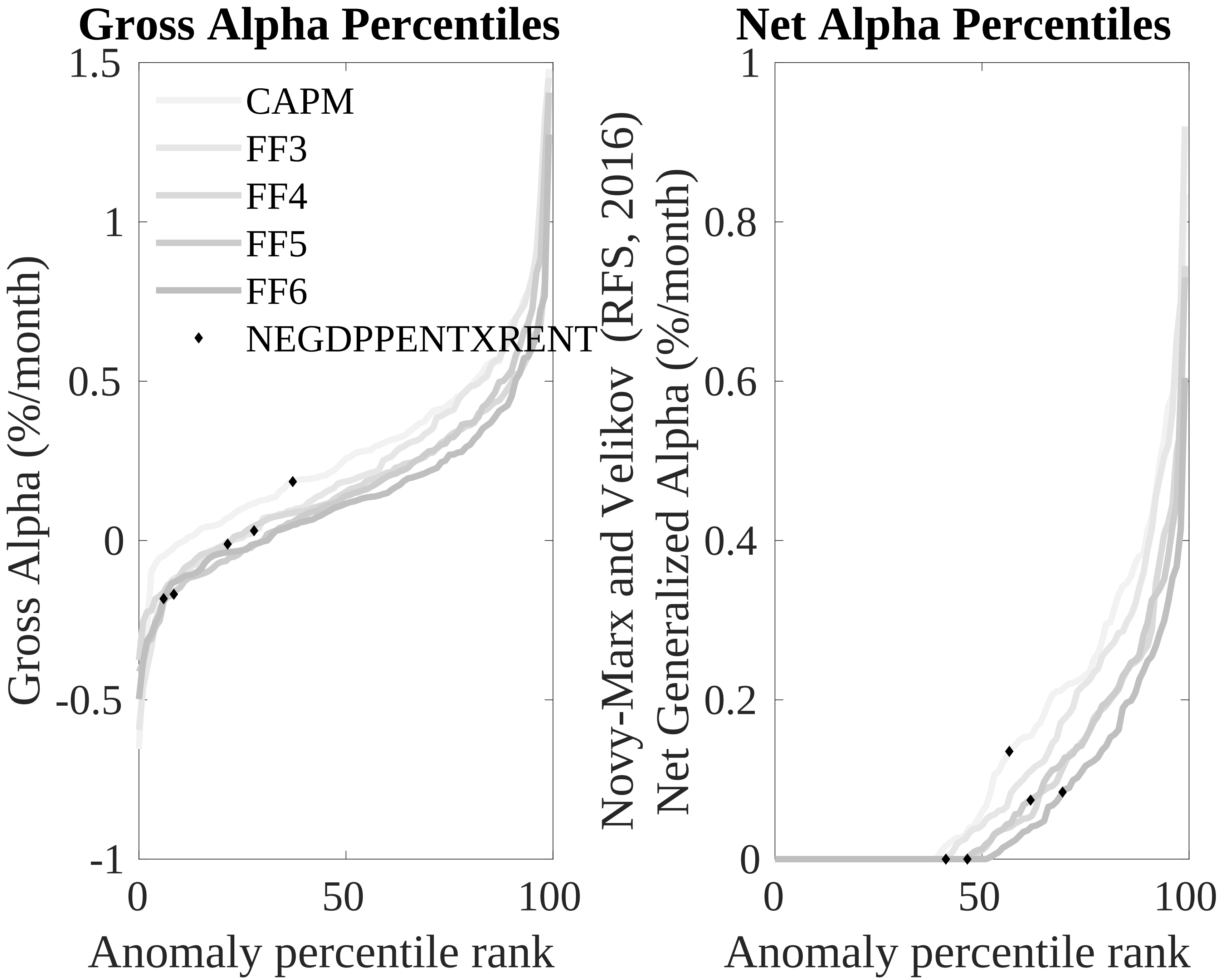
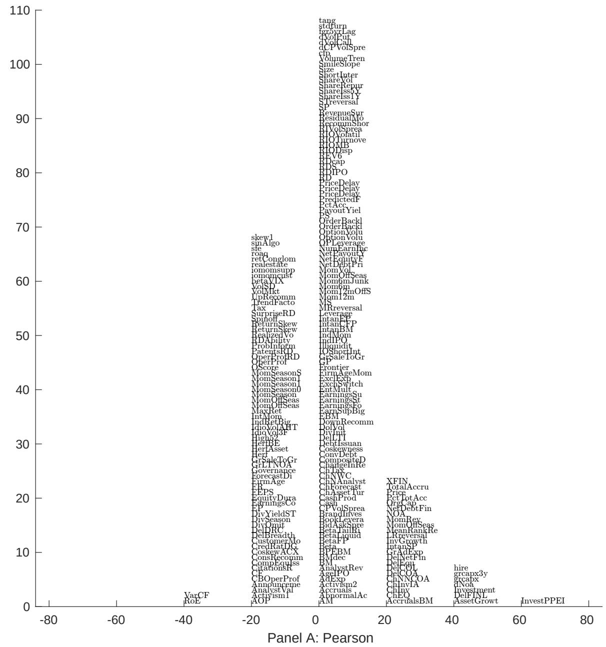


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ALI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDPPENTXRENT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



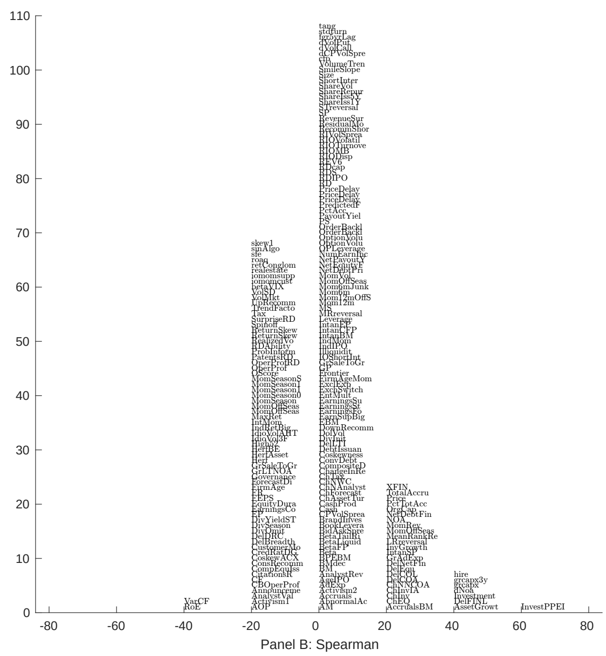


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with ALI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

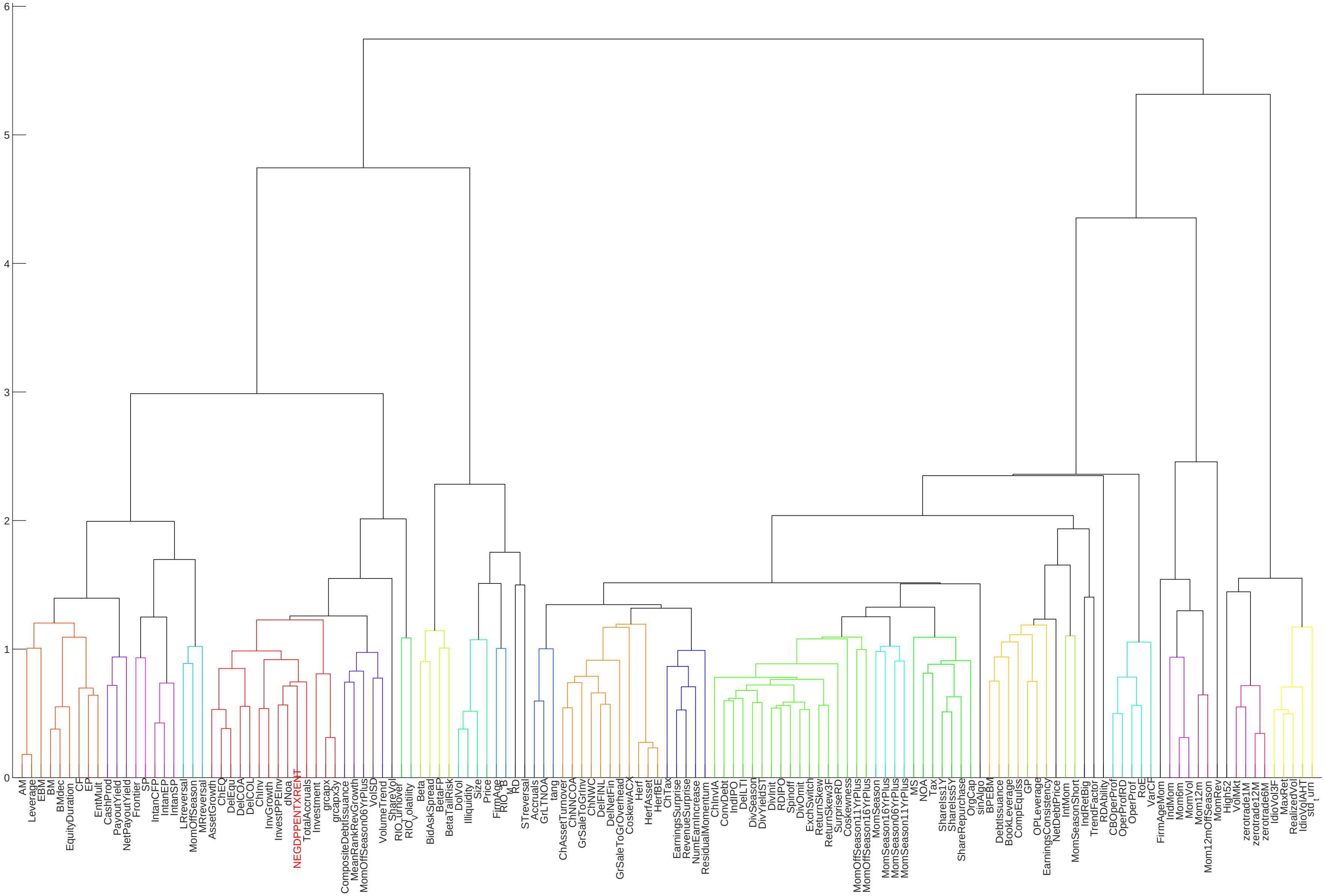


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

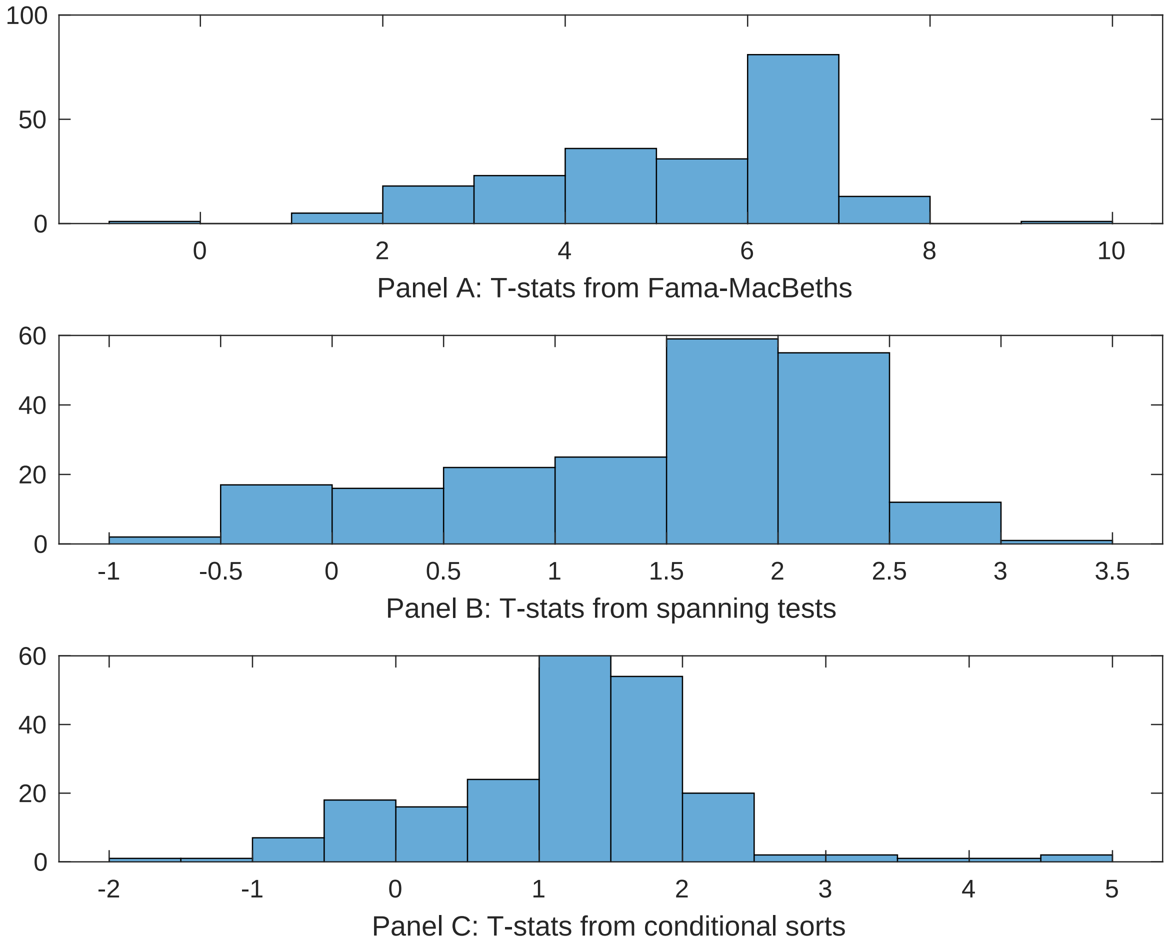


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ALI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ALI} ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ALI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ALI} ALI_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.14	0.13	0.13	0.14	0.13	0.14
	[6.02]	[6.56]	[5.92]	[5.87]	[6.14]	[5.51]	[6.05]
ALI	0.33	0.49	0.64	0.44	0.40	0.71	0.25
	[2.54]	[3.70]	[4.79]	[3.36]	[3.04]	[5.84]	[1.88]
Anomaly 1	0.14						-0.13
	[7.37]						[-0.54]
Anomaly 2		0.12					0.29
		[7.09]					[1.15]
Anomaly 3			0.55				0.12
			[5.97]				[0.85]
Anomaly 4				0.11			0.38
				[7.07]			[1.89]
Anomaly 5					0.93		0.46
					[7.71]		[2.50]
Anomaly 6						0.80	0.11
						[5.49]	[0.81]
# months	732	727	727	720	732	713	708
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ALI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ALI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.18	-0.15	-0.17	-0.24	-0.19	-0.12	-0.18
	[-2.36]	[-1.89]	[-2.09]	[-2.99]	[-2.24]	[-1.45]	[-2.44]
Anomaly 1	50.95						39.19
	[14.42]						[9.92]
Anomaly 2		38.64					23.50
		[9.77]					[3.58]
Anomaly 3			33.98				-1.70
			[8.45]				[-0.26]
Anomaly 4				37.91			14.01
				[8.23]			[2.89]
Anomaly 5					17.24		3.75
					[3.40]		[0.75]
Anomaly 6						25.91	2.28
						[5.62]	[0.52]
mkt	10.27	10.32	10.61	11.08	10.92	11.21	10.04
	[5.70]	[5.36]	[5.44]	[5.66]	[5.38]	[5.58]	[5.75]
smb	27.93	22.39	25.82	30.37	28.59	31.15	24.07
	[10.70]	[7.77]	[9.02]	[10.74]	[9.65]	[10.71]	[8.93]
hml	3.32	1.45	2.69	5.92	8.37	4.26	-1.13
	[0.95]	[0.38]	[0.70]	[1.56]	[2.13]	[1.07]	[-0.32]
rmw	24.82	22.62	22.08	25.42	24.87	25.59	23.92
	[7.06]	[6.02]	[5.77]	[6.66]	[6.28]	[6.53]	[6.99]
cma	22.04	40.42	42.47	32.86	40.34	38.33	1.71
	[3.80]	[6.93]	[7.15]	[5.03]	[4.80]	[5.46]	[0.22]
umd	1.92	-0.26	-0.48	1.22	2.56	0.57	0.35
	[1.07]	[-0.14]	[-0.25]	[0.63]	[1.27]	[0.28]	[0.20]
# months	732	728	728	732	732	713	709
$\bar{R}^2$(%)	49	42	41	40	36	38	53

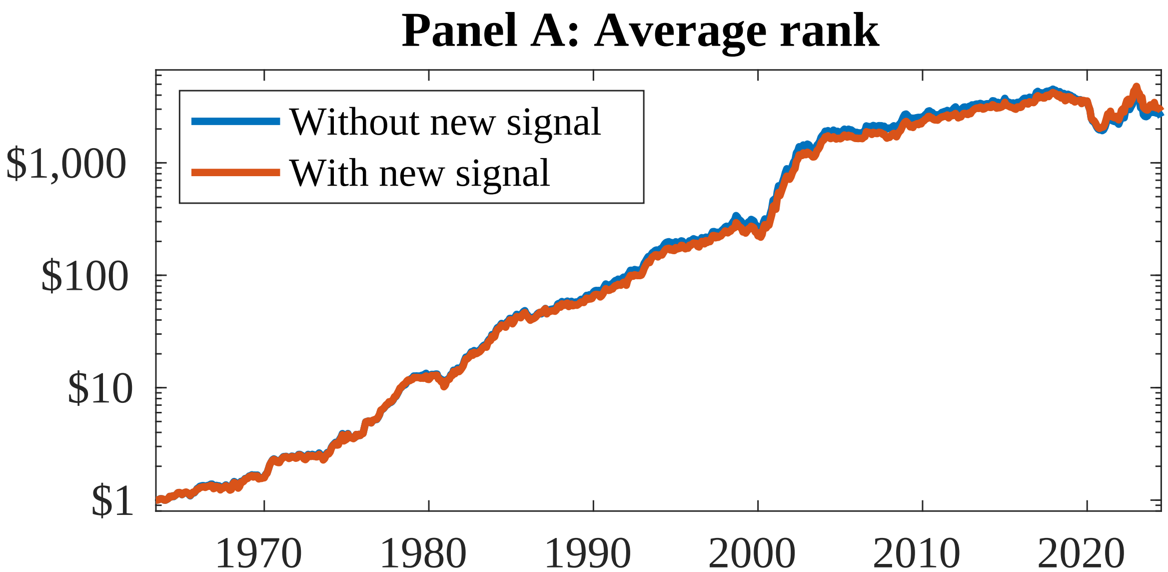


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ALI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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